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(1) Research Program/Project/Study Title:		Banana Pith Siomai with Saluyot (<i>Corchorus olitorius</i>) Leaves
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Sigma
		Province:	Capiz
		Region:	VI
(5) Classification Research: Development: ☐ Basic ☒ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Agriculture/crops		(8) Discipline/Field of Specialization Food	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Abstract

A growing need for delicious, nutritious and convenient food encourages the demand for siomai. The classic siomai is never never outdated because its filling can be any ingredients to satisfy difference dietary needs. The objectives of this study are to determine the sensory qualities of Banana Pith Siomai with Saluyot (*Corchorus olitorius*) Leaves in terms of appearance, aroma, taste, texture; determine the general acceptability of the product and determine if there are significant differences in the sensory qualities and general acceptability of the products among treatments. This study was conducted at Capiz State University Sigma Satellite College, Sigma, Capiz during the School Year 2023-2024. This experimental study utilized CRD as research design with five Treatments replicated three times evaluated by 12 evaluators using nine-point Hedonic scale. The statistical tools used were mean and ANOVA.

Results of the study revealed that Treatment A (control) got the highest mean score in terms of appearance, aroma, taste, texture, and general acceptability. Further revealed that there is a significant difference among treatments in terms of its general acceptability in favor of Treatment A which is also comparable to Treatment B.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Sensory qualities, general acceptability, nine-point Hedonic scale, analysis of variance

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Introduction

Anchored on the republic Act 10611, an act strengthen the food safety regulatory system in the country to protect consumer health and facilitate market access of local foods and food products and for other purpose, the researcher wanted to utilized indigenous vegetables found in the locality.

A growing need for delicious, nutritious and convenient food encourages the demand for siomai. The classic siomai is never never outdated because its filling can be any ingredients to satisfy difference dietary needs. It is a traditional Chinese dumpling and has made its way to the heart of the Filipino's as evidenced by the hundreds of stalls, eateries, and restaurants who serve them. It Basically makes use of ground pork, shrimps, and mushrooms as a main ingredients (Merano, 2023).

Banana Pith are botanically a part of the Musa genus and are actually a flower stalk of a large, herbaceous plant belonging to the Musaceae family. All parts of the banana plant are edible, and though the fruits are the most commonly consumed portion of the plant, the leaves and stalks are also used for a variety of culinary applications in Asia (Specialty Produce, 2015).

Banana stems are rich in potassium and magnesium. Potassium and magnesium can prevent kidney stones to a great extent. The magnesium present in banana stem readily combines with the excess oxalates, thereby preventing the formation of calcium oxalate stones in the kidneys. It is also enriched in fibers, iron, calcium, and other beneficial substances. Banana stem juice also wonderfully reduces and cures kidney stones. It cures calcium oxalate and uric acid kidney stones by dissolving them (Pristyn Care Team, 2022).

Saluyot (*Corchorus olitorius*) is an edible leafy vegetable that is a member of the genus *Corchorus*, classified under the subfamily Grewioideae of the family Malvaceae. Saluyot is widely found in tropical and subtropical areas from Asia to Africa valued as food and for its strong fiber. Saluyot leaves are very nutritious, it is rich in calcium, iron, protein, vitamin A, C and E, thiamin, riboflavin, niacin, folate, and dietary fibers. Saluyot is usually cooked as stew, forming a thick slimy syrup similar in consistency to okra usually taken with rice or other starchy staple. It can almost grow anywhere in the Philippines. Saluyot is a hardy plant that is resistant to pests and requires little care. It can be found in the wild as it can also be grown in a farm.

Vegetables play an important role in human nutrition. Most are low in fat and calories but are bulky and filling. They supply dietary fiber and are important sources of essential vitamins, minerals and trace elements.

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Particularly important are the antioxidant vitamins A, C and E. When vegetables are included in the diet, there is found to be a reduction in the incidence of cancer, stroke, cardiovascular disease and other chronic ailments

The American Frozen Food Institute (AFFI) predicts that consumers will continue to seek more nutritious, plant-based and foods that are more convenient, and the market for products that could meet the needs has a lot of room for growth and success. In addition to the HALAL Food Certification, various meat alternatives on the market, now there are chicken and seafood siomai available and for other purpose, the researcher wanted to utilized indigenous vegetables found in the locality like banana pith and saluyot leaves in making siomai.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Methodology

This study utilized experimental research design to determine the sensory qualities such as appearance, aroma, taste, texture; and general acceptability of utilizing banana pith and saluyot leaves in making siomai.

The experimental design used in the study was the single factor experiment with Completely Randomized Design (CRD) in which banana pith and saluyot leaves were utilized in making siomai. In this study, the banana pith and ground pork, were added in different proportions; Treatment A – 1,134g ground pork, Treatment B – 850.5g ground pork and 283.5g banana pith, Treatment C – 567g ground pork and 567g banana pith, Treatment D – 283.5g ground pork and 850.5g banana pith, and Treatment E – 1,134g banana pith. The powdered saluyot leaves will be added proportionately from Treatment B to E

Procedure in preparing banana pith includes cleaning, slicing, chopping, pounding, boiling, and draining. Procedure in preparing saluyot leaves includes cleaning, air-drying, and powdering. Procedure in making Treatment C banana pith siomai with saluyot leaves are the following: combining ground pork, banana pith, saluyot leaves powdered, minced shrimp, , gradually mixing together. Adding jicama, oil, seasoning, black pepper, minced onion, minced carrots, egg, seasoning, and spring onion. Scooping ½ T of pork mixture and place in wonton wrapper. Steaming for about 15 – 20 minutes. Cool, serve or package.

The arithmetic mean and Analysis of Variance (ANOVA) were the statistical tools used in analyzing and interpreting the data. The mean was used to determine the appearance, aroma, taste, texture, and general acceptability of the product. The ANOVA set at 0.05 alpha level was used to determine the significant differences on the sensory qualities and general acceptability of the products among treatments.

The study used a scorecard using a Nine-point Hedonic Scale. It looked into the four sensory qualities such as appearance, aroma, taste, and texture as the variables. These four sensory qualities will determine the general acceptability of banana pith patty with malunggay. The evaluators were 12 semi-skilled panelist composing faculty teaching food related subjects and Food Service Management students.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

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Results and Discussions

Table 1.0 Sensory qualities of Banana Pith Siomai with Saluyot Leaves.

Treatment	Mean	Verbal Interpretation
Appearance		
Treatment A	8.47	Extremely Attractive
Treatment B	7.53	Very Much Attractive
Treatment C	7.64	Very Much Attractive
Treatment D	7.61	Very Much Attractive
Treatment E	7.47	Very Much Attractive
Aroma		
Treatment A	8.28	Extremely Pleasant
Treatment B	7.72	Very Much Pleasant
Treatment C	7.72	Very Much Pleasant
Treatment D	7.47	Very Much Pleasant
Treatment E	7.33	Very Much Pleasant
Taste		
Treatment A	8.19	Liked Extremely
Treatment B	7.39	Liked Very Much
Treatment C	7.11	Liked Moderately
Treatment D	7.17	Liked Moderately
Treatment E	7.11	Liked Moderately
Texture		
Treatment A	8.17	Extremely Firm
Treatment B	7.39	Very Much Firm
Treatment C	7.42	Very Much Firm
Treatment D	7.06	Moderately Firm
Treatment E	7.22	Moderately Firm

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Sensory Qualities of Banana Pith Siomai with Saluyot Leaves

Table 1.0 deals with the sensory qualities of banana pith siomai with saluyot leaves in terms of appearance, aroma, taste, and texture. In terms of appearance, product in Treatment A (control) was found “Extremely Attractive” with the mean score of 8.47, while product in Treatment C (50% ground pork and 50% banana pith) M=7.64; Treatment D (25% ground pork and 75% banana pith) M=7.61; Treatment B (75% ground pork and 25% banana pith) M= 7.53; and Treatment E (100% banana pith) M= 7.47 were found “Very Much Attractive” respectively. It implies that among the Treatments evaluated, products in Treatment A was extremely attractive as rated by the evaluators.

Same Table shows that in terms of aroma, Treatment A was found “Extremely Pleasant” with the mean score of 8.28. Treatment B and C (M=7.72), Treatment D (M=7.47), and Treatment E (M=7.33), were found to be “Very Much Pleasant”. It implies that product in Treatment A got the highest mean rating and was rated as extremely pleasant.

With regards to taste, Table shows that Treatment A was found to be “Liked Extremely” with the mean score of 8.19. Treatment B (M=7.39) found to be “Liked Very Much” while, Treatments D (M= 7.17), Treatment C (M= 7.11), and Treatment E (M=7.11), were found “Liked Moderately” by the evaluators. It implies that among the treatments evaluated, product in Treatment A got the highest mean score and was liked very much by the panelist.

Same Table shows that in terms of texture, Treatment A was found “Extremely Firm” with the mean score of 8.17, while Treatment C (M=7.42) and Treatment B (M=7.39) were found to be “Very Much Firm”. Likewise, Treatment E with the mean score of 7.22 and Treatment D with the mean of 7.06 were found to be “Moderately Firm”. It implies that Treatment A got the highest mean score among other Treatments in terms of texture.

Table 2.0. General acceptability of Banana Pith Siomai.

Treatment	Appearance	Aroma	Taste	Texture	Grand Mean	Verbal Interpretation
Treatment A	8.47	8.28	8.19	8.17	8.26	Extremely Acceptable
Treatment B	7.53	7.72	7.39	7.39	7.51	Very Much Acceptable
Treatment C	7.64	7.72	7.11	7.42	7.47	Very Much Acceptable
Treatment D	7.61	7.47	7.17	7.06	7.33	Very Much Acceptable
Treatment E	7.47	7.33	7.11	7.22	7.29	Very Much Acceptable

General Acceptability of Banana Pith Siomai with Saluyot Leaves

Data in Table 2.0 revealed the general acceptability banana pith siomai with saluyot leaves. Results revealed that product from Treatment A got the highest mean of 8.26 verbally interpreted as “Extremely Acceptable”, while product in Treatment B (M=7.51); Treatment C (M=7.47); Treatment D (M=7.33); and Treatment E (M=7.29) were very much acceptable as rated by the evaluators. This indicated that Treatment A was the most acceptable product by the evaluators as indicated by its mean. However, The other treatment also were acceptable as evident by their means. It implies that using banana pith with the different formulation was also accepted to the evaluators.

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Table 3.0 Analysis of variance of the sensory qualities and general acceptability of Banana Pith Siomai with Saluyot Leaves.

Source of Variance	Sum of Squares	df	Mean Square	F	Sig.
Appearance					
Between Groups	7.215	4	1.804	1.174	0.332
Within Groups	84.519	55	1.537		
Total	91.733	59			
Aroma					
Between Groups	6.252	4	1.563	1.422	0.239
Within Groups	60.435	55	1.099		
Total	66.687	59			
Taste					
Between Groups	10.230	4	2.557	0.947	0.444
Within Groups	148.546	55	2.701		
Total	158.776	59			
Texture					
Between Groups	8.711	4	2.178	0.726	0.578
Within Groups	165.028	55	3.001		
Total	173.739	59			
General Acceptability					
Between Groups	7.621	4	1.905	1.194	0.032
Within Groups	87.749	55	1.595		
Total	95.370	59			

Differences on Sensory Qualities and General Acceptability of Banana Pith Siomai with Saluyot Leaves

Table 3.0 presents the Analysis of Variance on the sensory qualities such as appearance, aroma, taste, and texture and general acceptability of banana pith siomai with saluyot leaves.

In terms of appearance, data show that there is no significant differences among Treatments ($F=1.174$; $p>0.05$). It implies that appearance of the product among treatment were the same.

When aroma was considered in the grouping, the same Table shows no significant difference existed among Treatments ($F=1.422$; $p>0.05$).

In terms of taste, it can be noted no significant differences among Treatment ($F=0.947$; $p>0.05$) implying that taste of the products were the same as evaluated by the panelist.

For the texture of the product Table shows no significant difference existed among Treatments ($F=0.726$; $p>0.05$). It implies that texture of the product in all treatments did not varies.

Interestingly, in terms of general acceptability of the products, data show a significant difference existed among Treatments ($F=1.194$; $p<0.05$). It implies that acceptability of the product among treatment differed in favor of Treatment A as evaluated by the panelist. It can said that treatment B was also comparable to Treatment A.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

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Conclusions and Recommendations

It can be concluded that in terms of appearance, aroma, taste, and texture, treatment A (control) got the highest mean followed by treatments B and C. It is recommended that adding portion of banana pith in making siomai not only making its sensory qualities improve but it add a promise of pack of health benefits.

It also concluded that in terms of general acceptability, Treatment A followed by treatment B were accepted as evaluated by the panelist. Since treatment B (75% ground pork and 25% banana pith) may comparable to treatment A in terms of acceptability, this proportion may use in making siomai since banana is abundant in locality.

There is a significant difference in the general acceptability of the product among treatments in favor of treatment A. Likewise, Treatment B was found comparable to the control, it may recommend that adding banana pith in the formulation of making siomai is a great alternative to pork which create a new version that is not familiar yet expected to be satisfying.

The accepted Treatment may be tested for its self life under different temperature conditions, proximate, and microbial analysis.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Banana Pith Siomai

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

Banana Pith Siomai

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

Research Office Capiz State University Sigma Satellite College

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

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